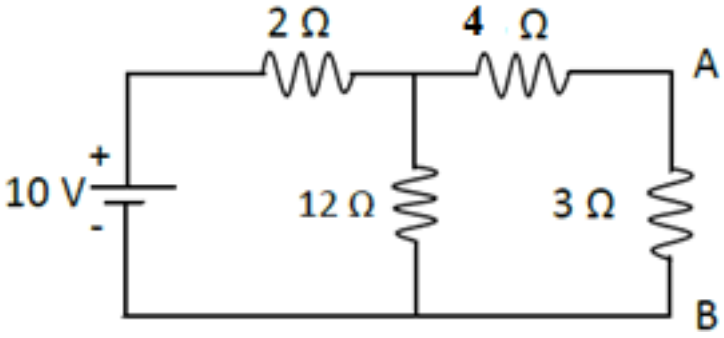
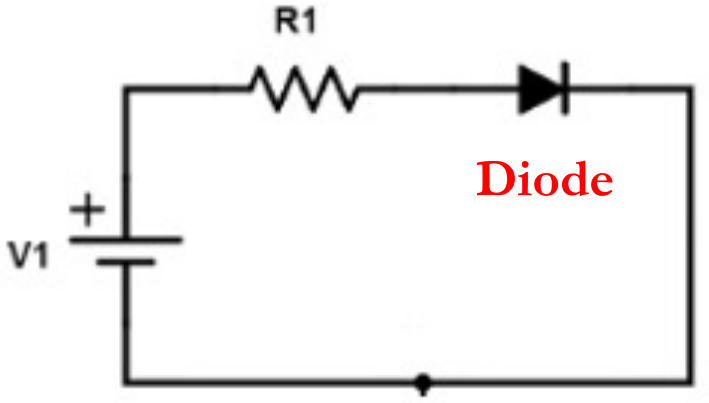
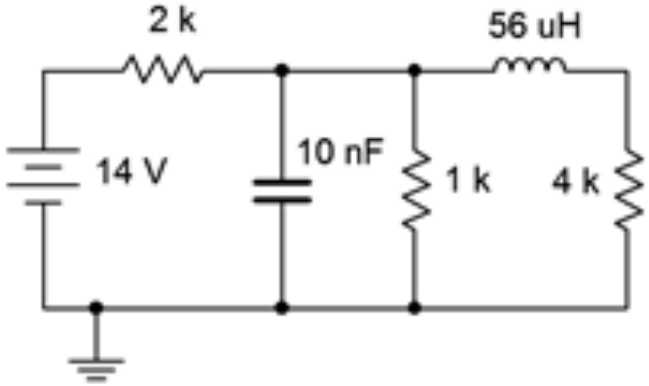
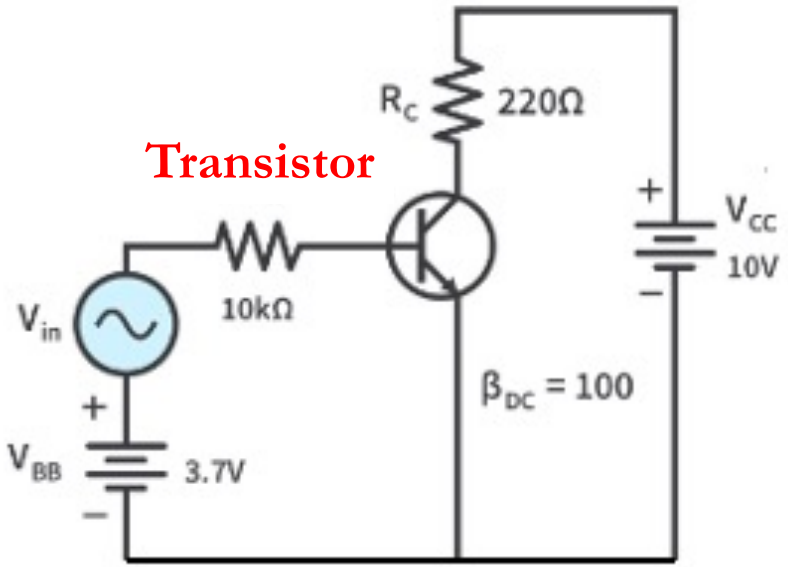
Course Introduction

Electronic Circuits

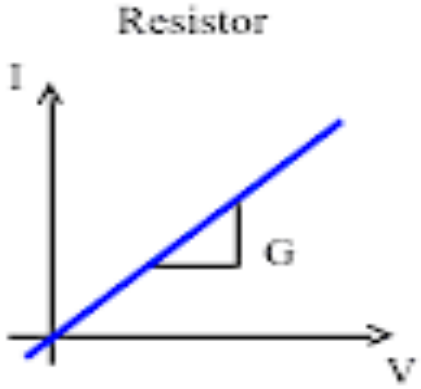




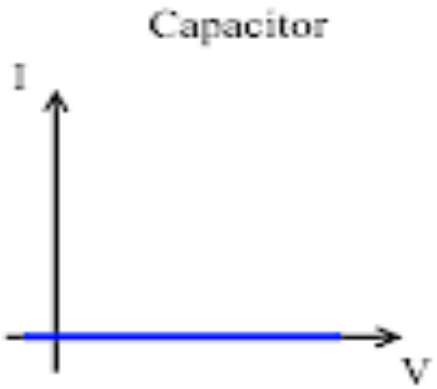




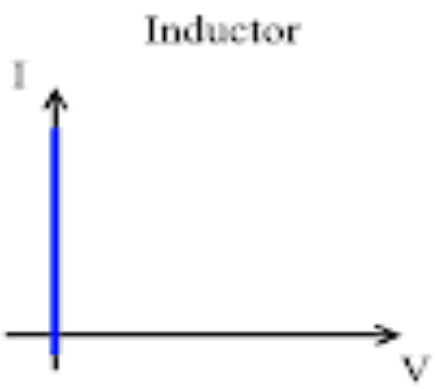
Passive Electronic Components
Capacitors and Inductors



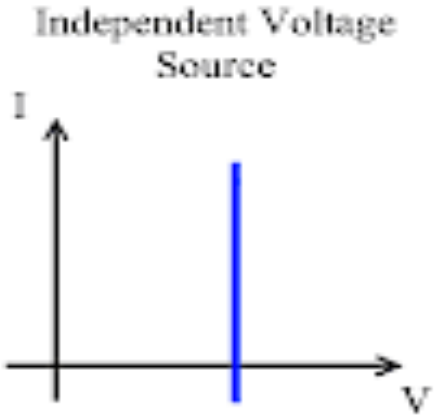
I = GV



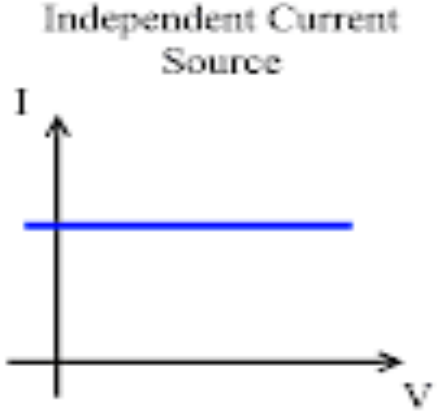
I = 0 for all V



V = 0 for all I

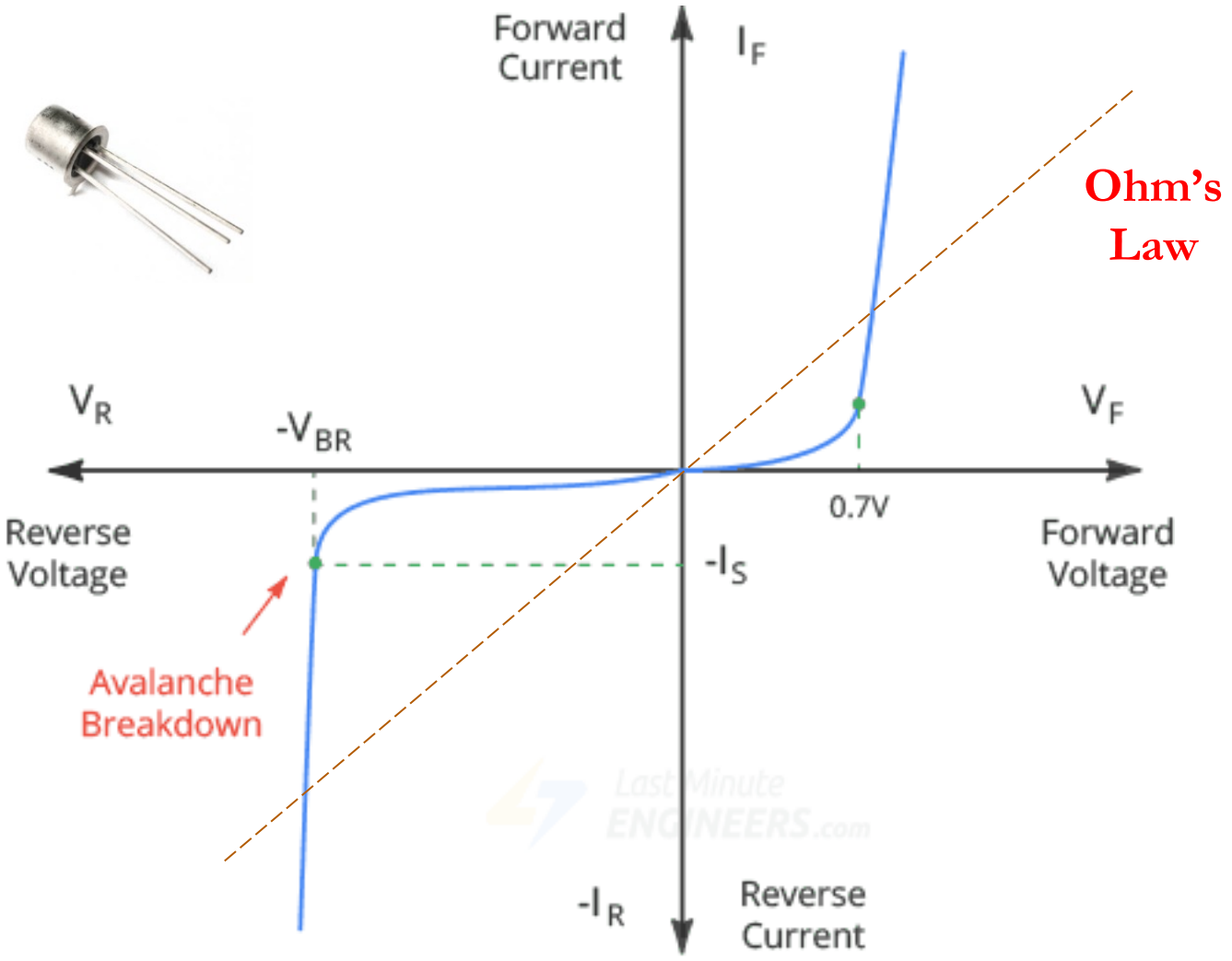


V constant for all I

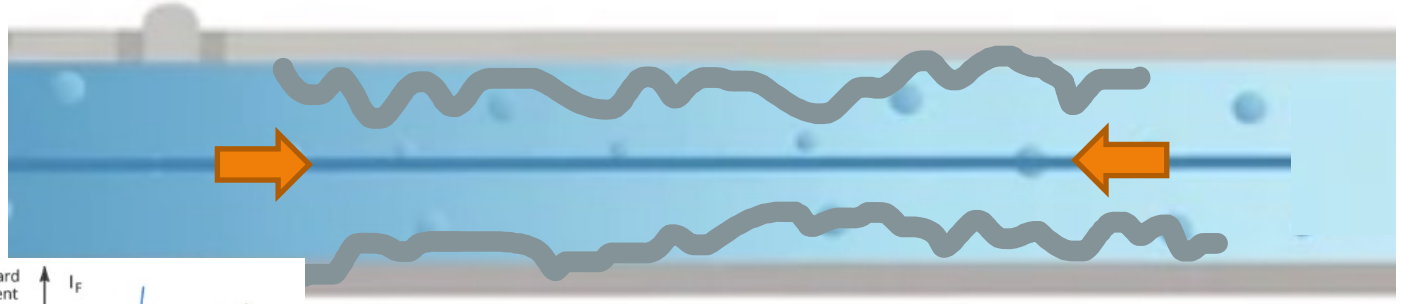


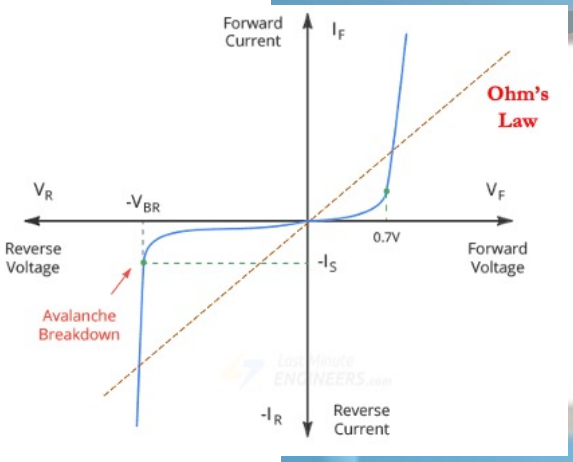
I constant for all V

Active Electronic Components

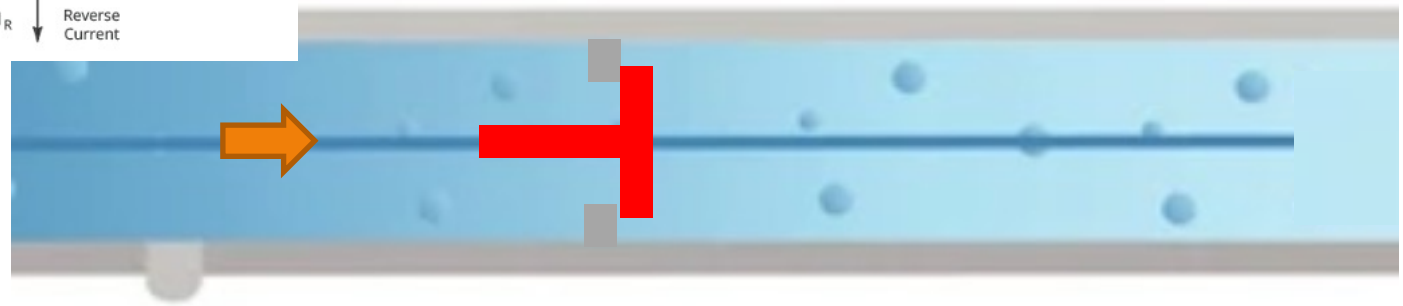


Active Electronic Components





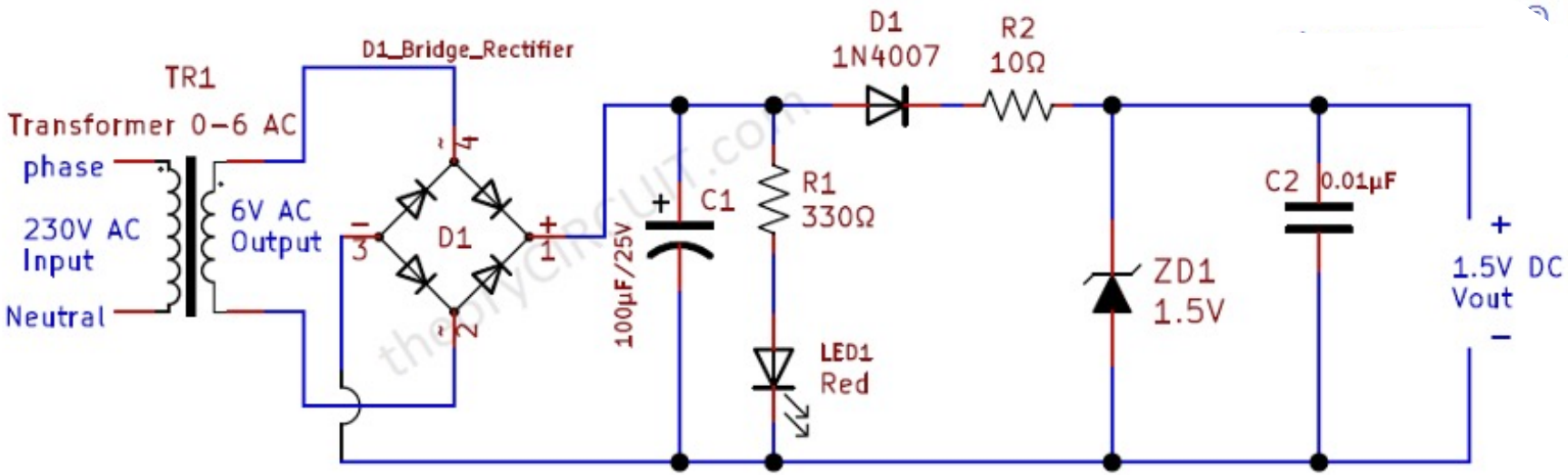
Resistor

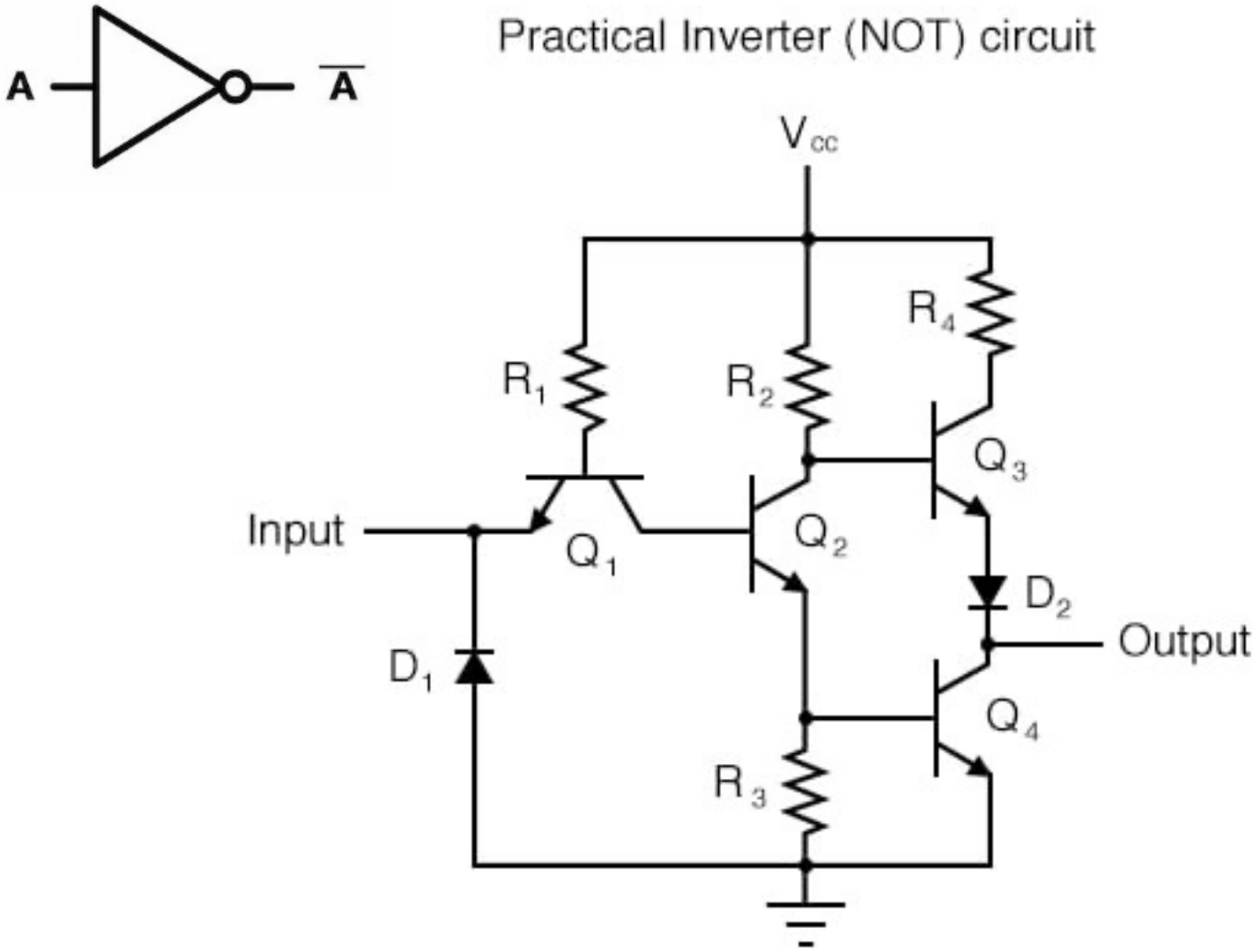


Diode



1.5V DC Zener Diode Regulator Circuit

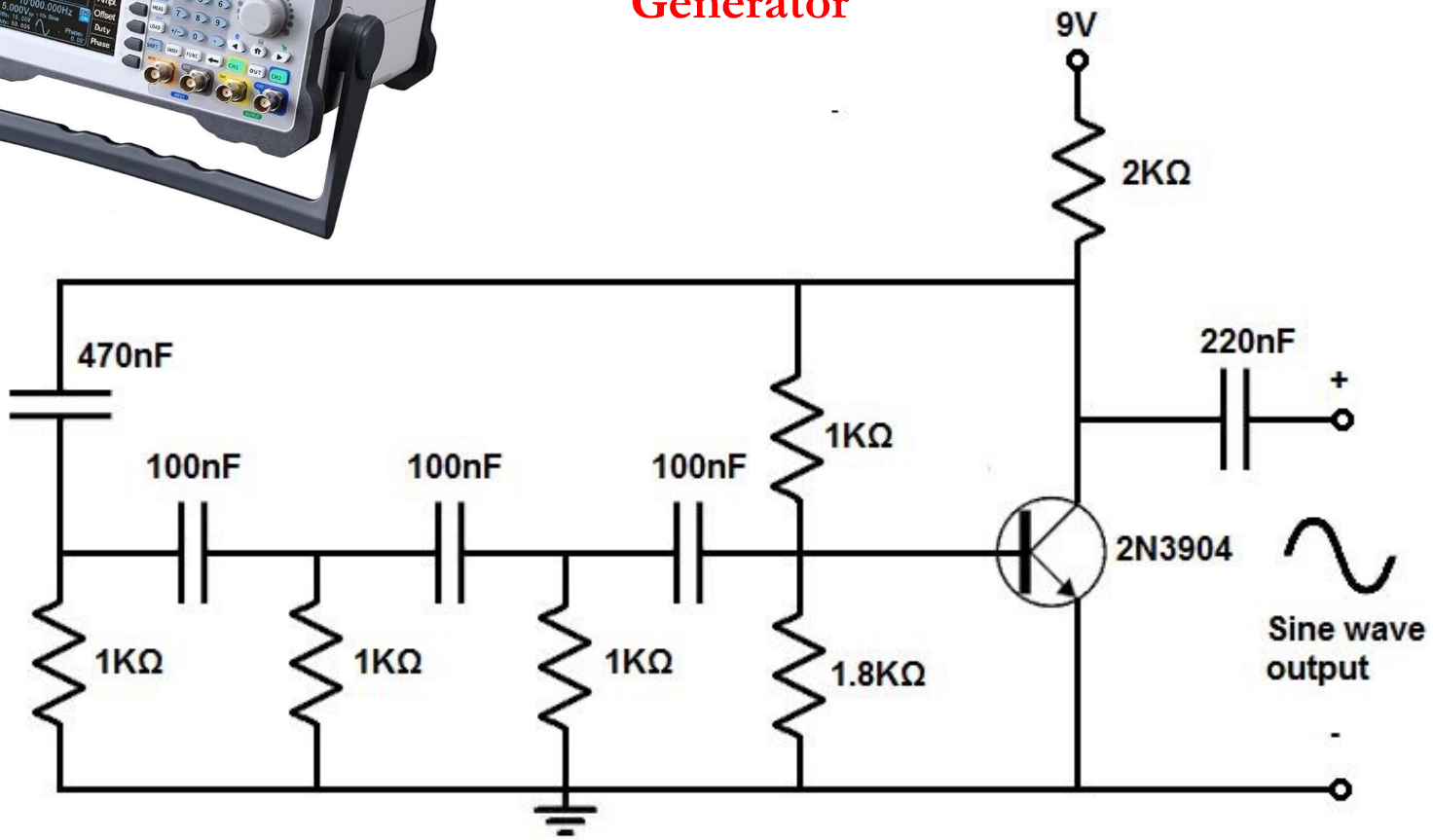


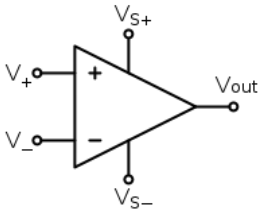

A
$\overline{A}$
Practical Inverter (NOT) circuit
V_{cc}
R_1
R_2
R_4
Q_3
Input
Q_1
Q_2
D_2
Output
D_1
Q_4
R_3

Electronic Circuits



Signal Generator



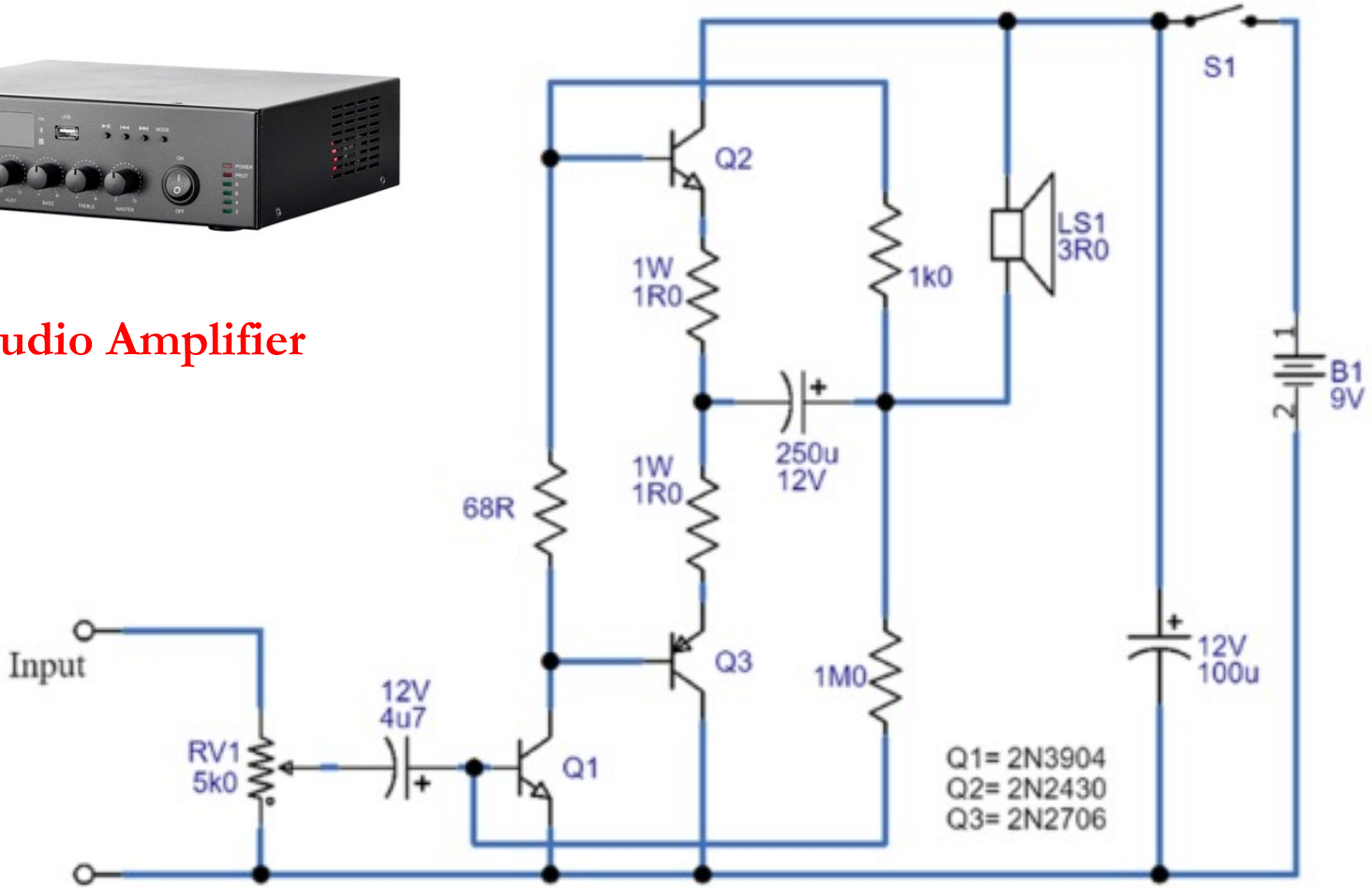


Oprational Amplifier

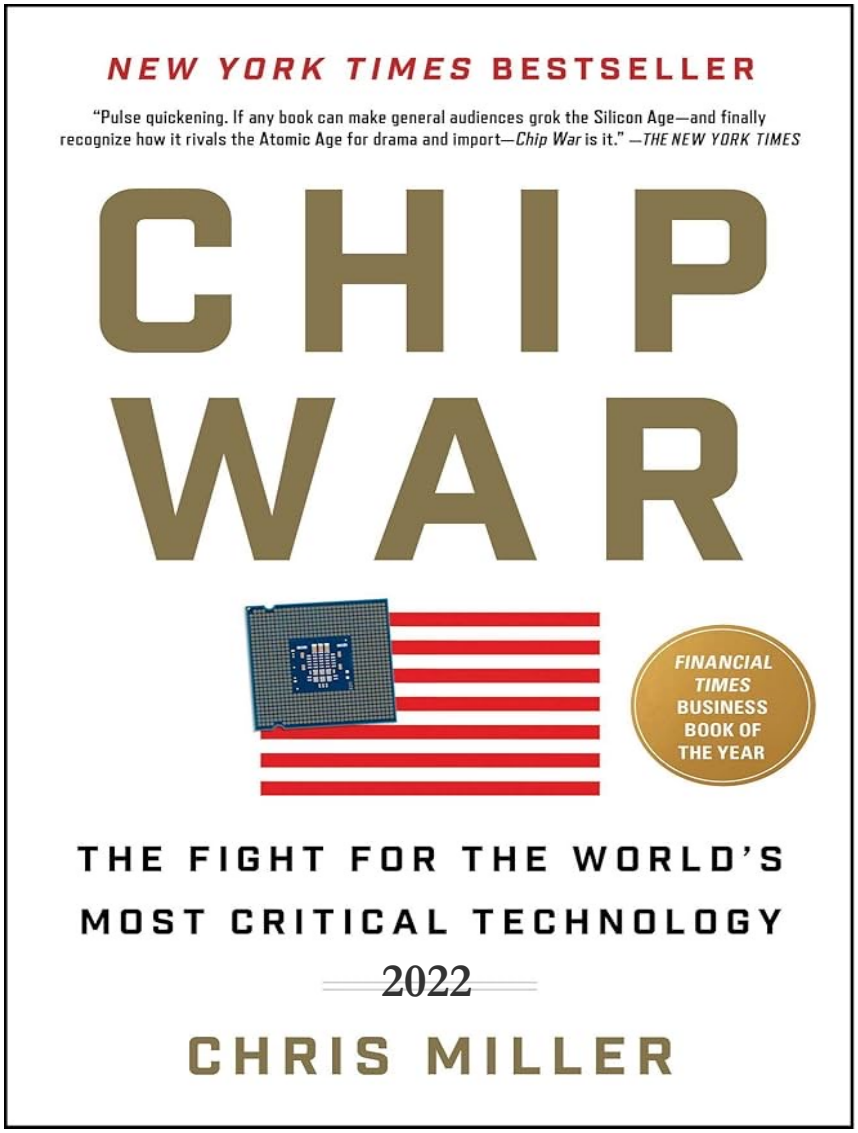
Electronic Circuits



Audio Amplifier



Electronic Technology

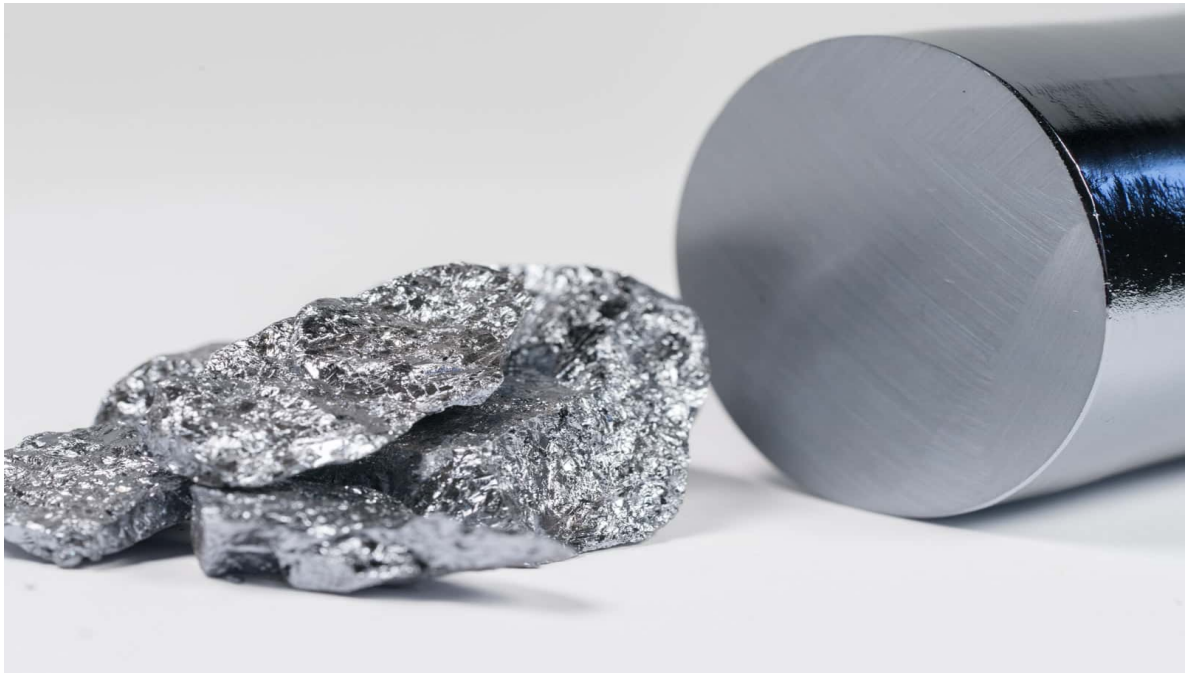


1950's:
Transistor: When the world-changing invention transistor was invented, The New York Times buried the story on page 46. Bell Labs scientists John Bardeen, Walter Brattain, and William Shockley won the **1956 Nobel Prize** in Physics for their invention of the transistor.

Semiconductor/Chips: After World War II, Texas Instruments (TI) executives realized that electronics expertise will be useful for military systems and they hired people like Jack Kilby. The eight people who left Shockley's Lab are widely credited with the creation of Silicon Valley. One of the eight Eugene Kleiner would go on to found Kleiner Perkins, Gordon Moore who went on to run Fairchield's R&D and also the one who coined the concept Moore's Law (The number of transistors in an **integrated circuit** (IC) doubles about every two years. It became Moore's Law).

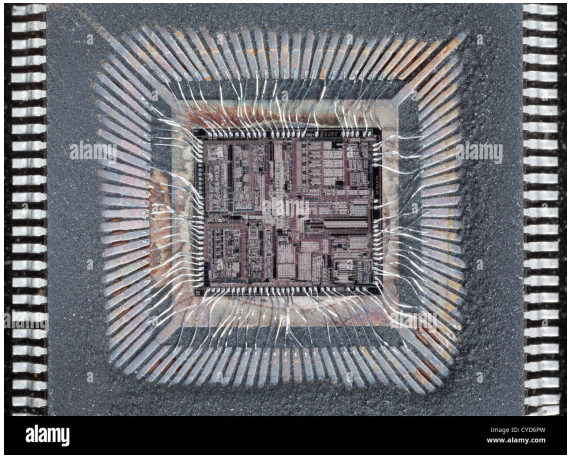
Sputnik, Yuri Gagarin and **Space** Sputnik, the first artificial Earth satellite was launched by the Soviet Union in 1957 and four years later Russian cosmonaut Yuri Gagarin became the first person in space. This is around the same time that Noyce and Moore founded Fairchild Semiconductor.

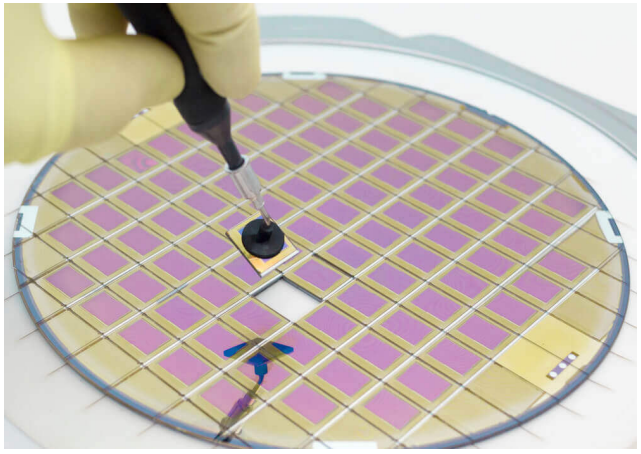
Electronic Technology











Syllabus

Nano Photonics and Plasmonics Research

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Office Hours Mon Wed Thu: 13:00-14:00	Phone ---	Email mohammad.alsunaidi@marmara	
Course Information	Text: Sedra/Smith, *Microelectronic Circuits 5th Ed.*	Grading	Course Work 15%	Lab Work 15%	Mid-term Exam 30%	Final Exam 40%

Week	Events	Topics to cover	Ch	Sec	Lab Activity	Exercises
1 Feb 17-21	---	**Introduction to Physics of Semiconductors.** Conduction in Semiconductors, N-type and P-type Semiconductors, The PN junction.	3	7	NL	(3) 112,113,114,116,118.
2 Feb 24-28	---	**Diodes.** Introduction, The Ideal diode, Terminal characteristics of the Diode.	3	1-2	NL	(3) 2,3,4,6,9,10.
3 Mar 3-7	Course drop/add Week	**Diode Circuit Analysis.** Diode Models, Graphical and Analytical Diode Circuit Analysis.	3	3	Experiment (1) LTSpice	(3) 17,20,22,25,26, 40,41,46,47.
4 Mar 10-14	---	**Diode Applications.** The Zener diode. Half and Full-wave Rectifiers, Limiting and Clamping Circuits, Voltage Doublers. Other Diode types.	3	4-6, 8	Experiment (2) Diode ckt Simulation	(3) 64,65,66,71,76,78, 82,93,98,105.
5 Mar 17-21	---	**MOS Field-Effect Transistors (MOSFETs).** Device Structure and Modes of Operation. Types, Symbols and Conventions, Current-voltage (IV) Characteristics.	4	1-2	Experiment (3) Basic Diode Circuits	(4) 4,6,12,15,19,23.
6 Mar 24-28	Quiz 1 March 25 (In-class)	**MOSFET DC Circuits.** Biasing and Circuit DC Analysis.	4	3	Experiment (4) Diode Rectifier Circuit	(4) 35,36,37,41,42,44,47.
7 Apr 2-4	Mar 29 - Apr 1 Holidays	---	---	---	NL	---
8 Apr 7-11	---	**MOSFET Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CS, CG and CD).	4	4-6	Experiment (5) MOSFET DC Circuits	(4) 63,69,74,75,77.
April 14-20	**Mid-term Exams period**					
10 Apr 21-25	---	**MOSFET Amplifiers.** Single-stage Amplifiers (CS, CG and CD).	4	7	NL	(4) 81,85,88.
11 Apr 28-May 2	Labor Day: May 1	**Bipolar Junction Transistors (BJTs).** Structure and Modes of Operation, Types, Symbols and Conventions, Current-voltage (IV) Characteristics.	5	1-2	Experiment (6) MOSFET Amplifiers	
12 May 5-9	---	**BJT DC Circuits.** Biasing. BJT as a switch.	5	4	NL	(5) 3,4,7,13,20,31,39.
13 May 12-16	Midterm Make-up (Excused)	**BJT Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CE, CB and CC).	5	6	Experiment (7) BJT Characterization	(5) 69,71,72,74,79,82, 84,86.
14 May 19-23	Quiz 2 May 20 (In-class)	**BJT Amplifiers.** Single-stage Amplifiers (CE, CB and CC).	5	7	NL	(5) 104,106,107,115, 116,124.
15 May 26-30	---	**BJT Amplifiers.** Single-stage Amplifiers (CE, CB and CC).	5	7	Experiment (8) BJT Amplifiers	(5) 130,131,134.
16 June 2-4	June 5-9 Holidays	**CMOS Logic Circuits.** Overview, the CMOS Inverter, Other CMOS Logic circuits, CMOS transistor sizing	10	1-3	---	(10) 18,25,28.
17 June 10-13	Project submission Week	**Review**	---	---	---	---
June 16-29	**Final Exams period**	**June 30 - July 13 (Re-sit Final Exams period)**				

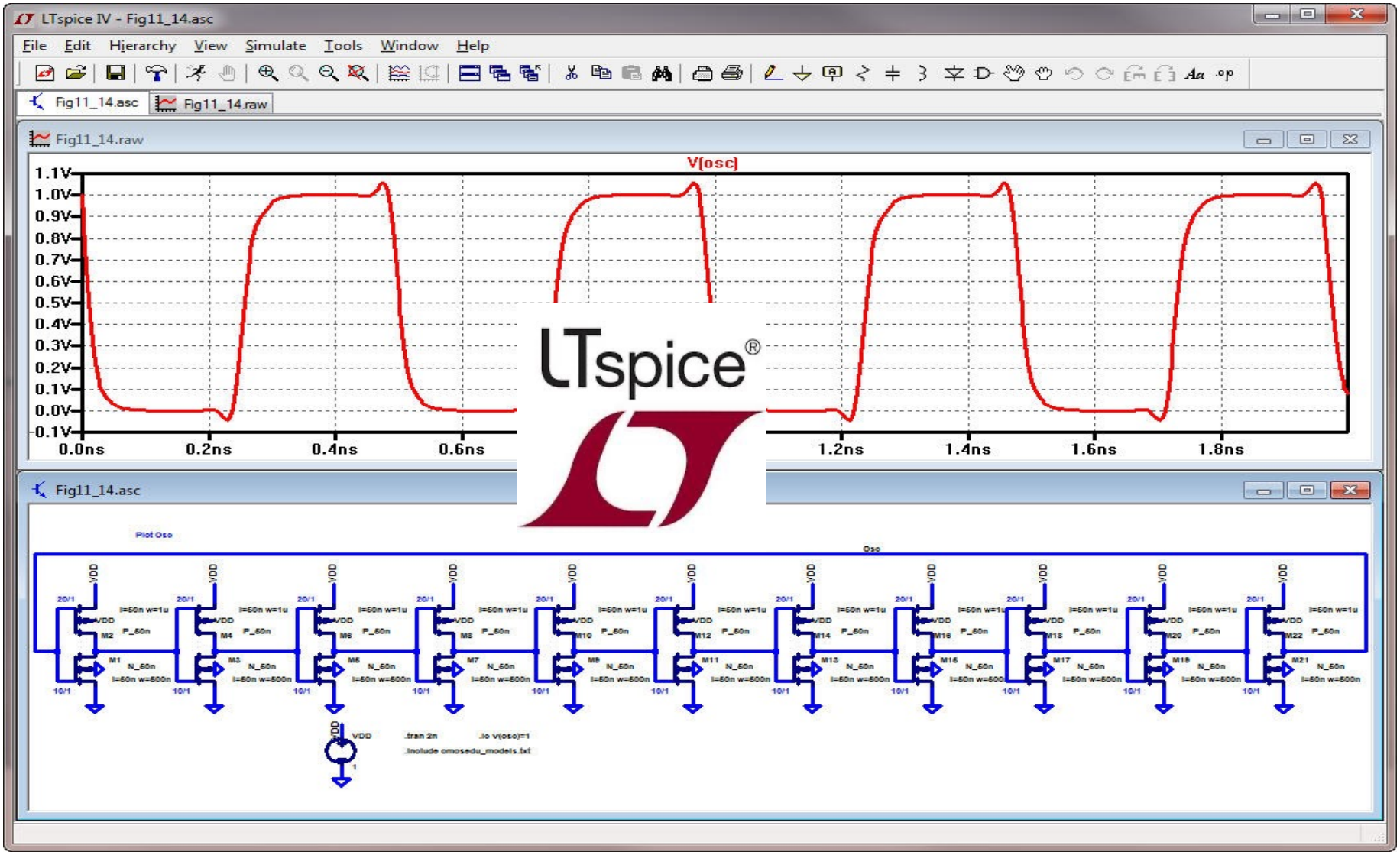
Pre-requisite Material

Physics II: Charges, Electric field.

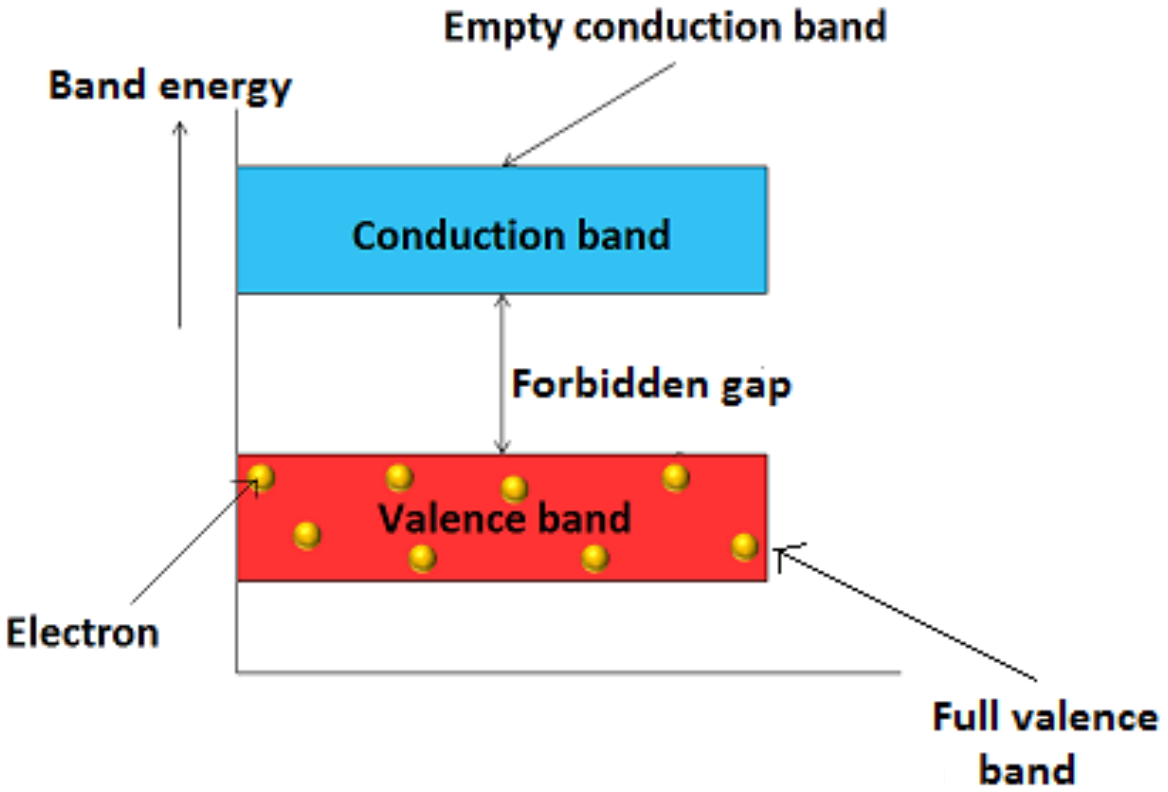
Math I: Simple integration and differentiation.

Circuits I: A lot of circuit analysis, including
the concepts of voltage, current and power.
Ohm's law
KVL and KCL equations.
Voltage and current dividers,
Thevenin's theorem,
Dependent sources, etc.

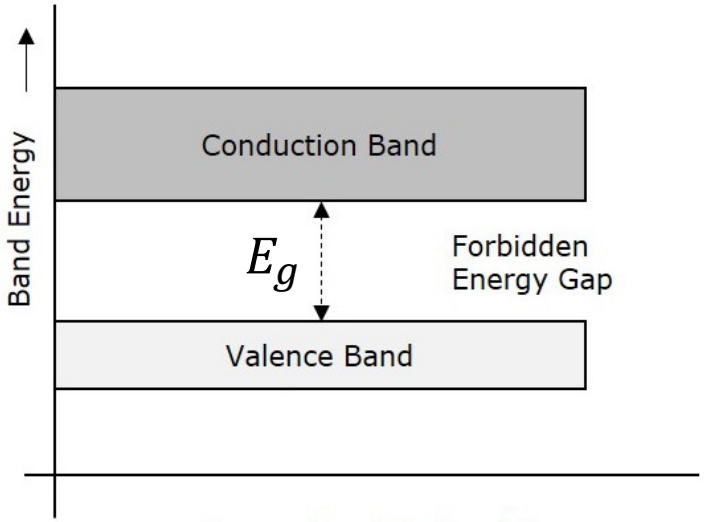
Simulation Software: LTspice



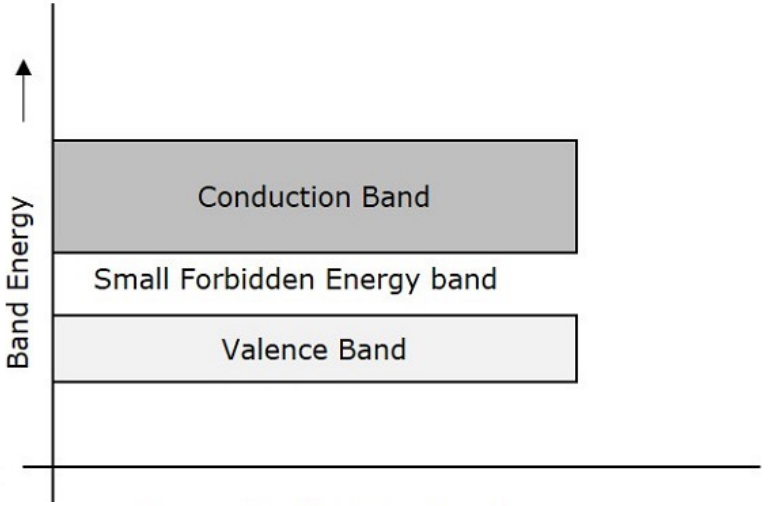
Energy Bands: Crystals



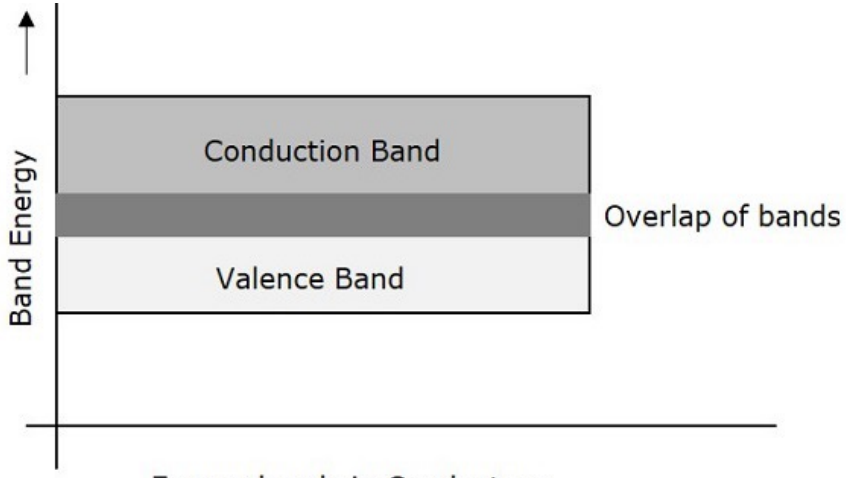
Classification of Materials



Energy bands in Insulators



Energy bands in Semiconductors



Energy bands in Conductors

Examples

Diamond $E_g = 6$ eV

Silicon $E_g = 1.12$ eV

Copper $E_g = 0$ eV

Common Semiconductors

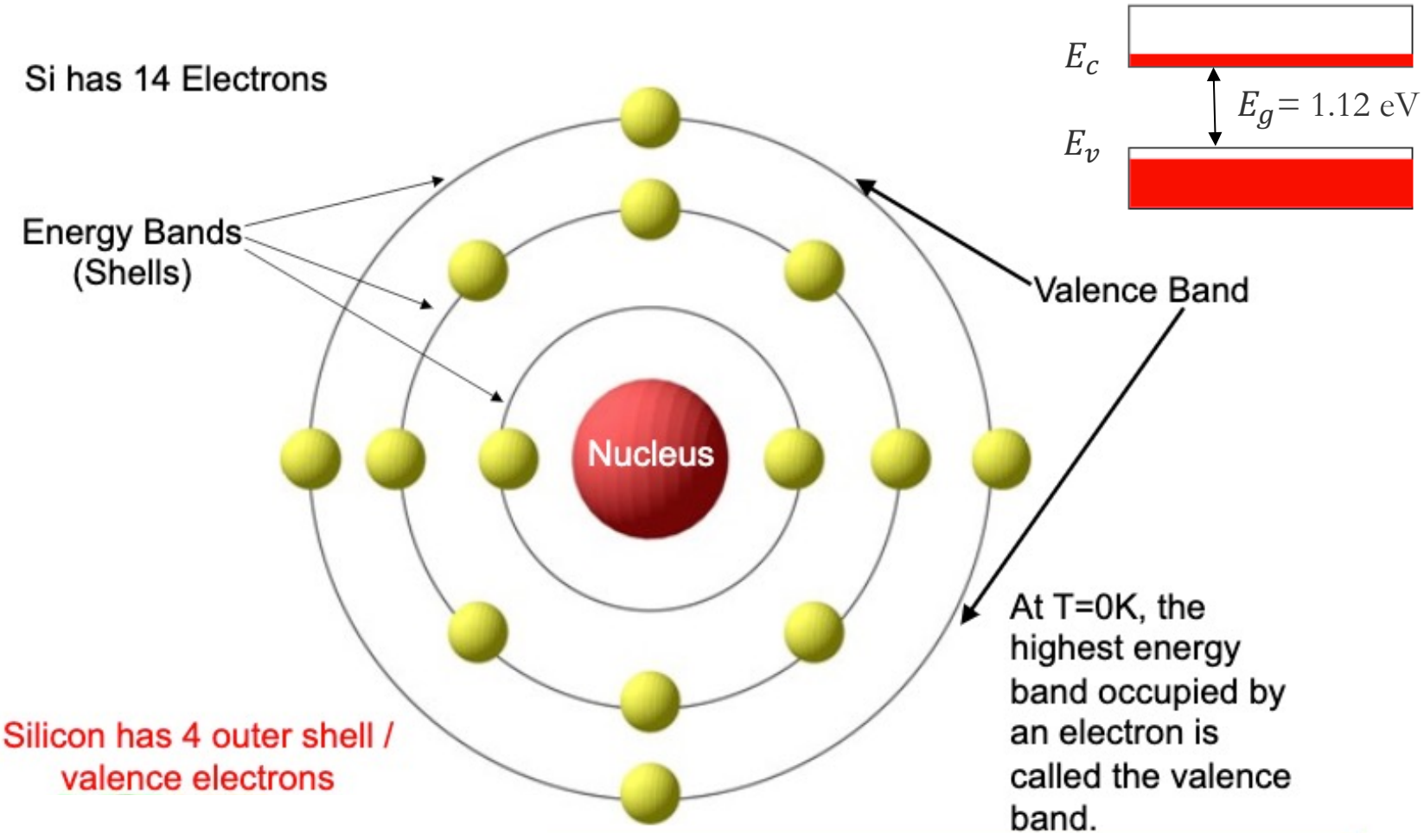
Semiconductor	Bandgap Energy E_G (eV)
Carbon (diamond)	5.47
Silicon	1.12
Germanium	0.66
Tin	0.082
Gallium arsenide	1.42
Gallium nitride	3.49
Indium phosphide	1.35
Boron nitride	7.50
Silicon carbide	3.26
Cadmium selenide	1.70

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

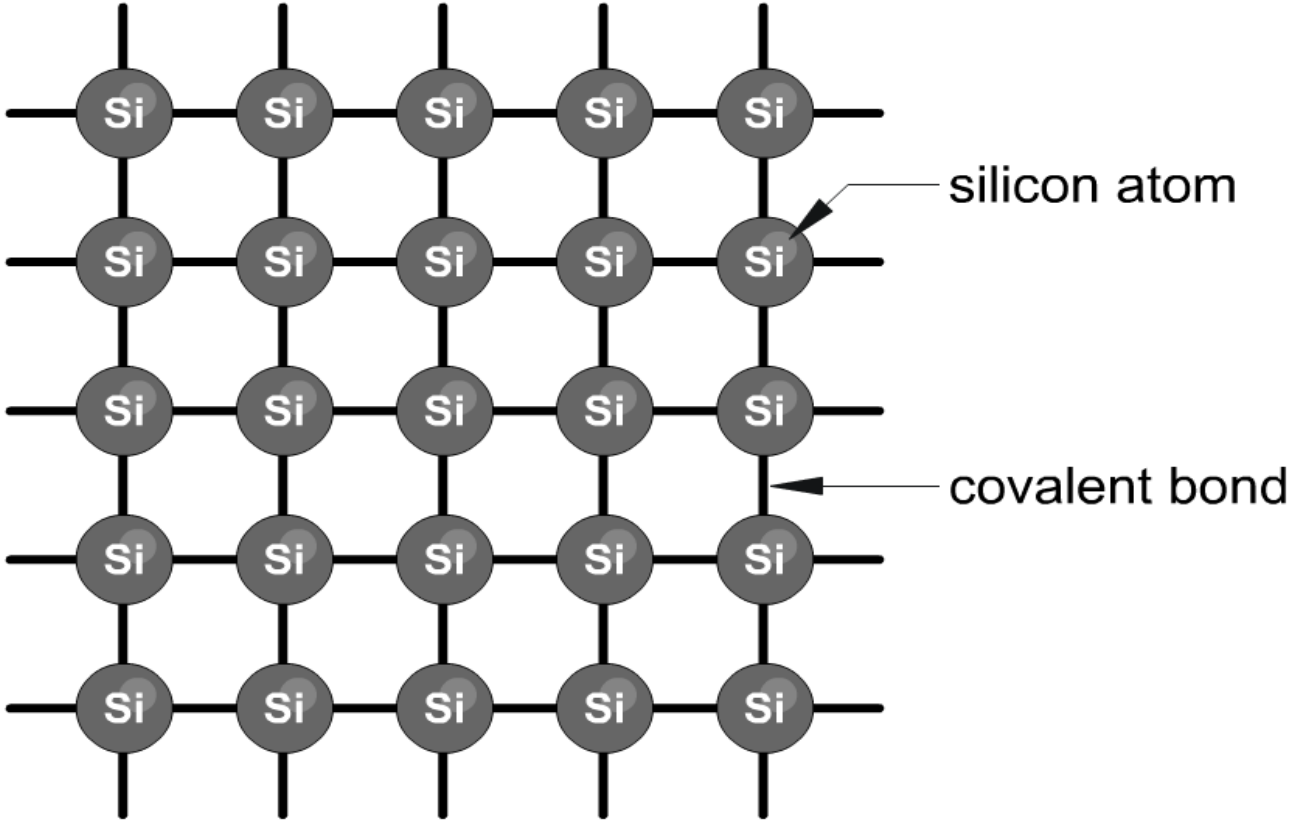
Example: Silicon



The Silicon Atom



Covalent Bonding in Pure Si



How many *free* electrons?

Carrier Count: Intrinsic Si

The density of free electrons is given by

$$n_i^2 = N_c N_v \, e^{-(E_c - E_v)/kT} = N_c N_v e^{-E_g/kT}$$

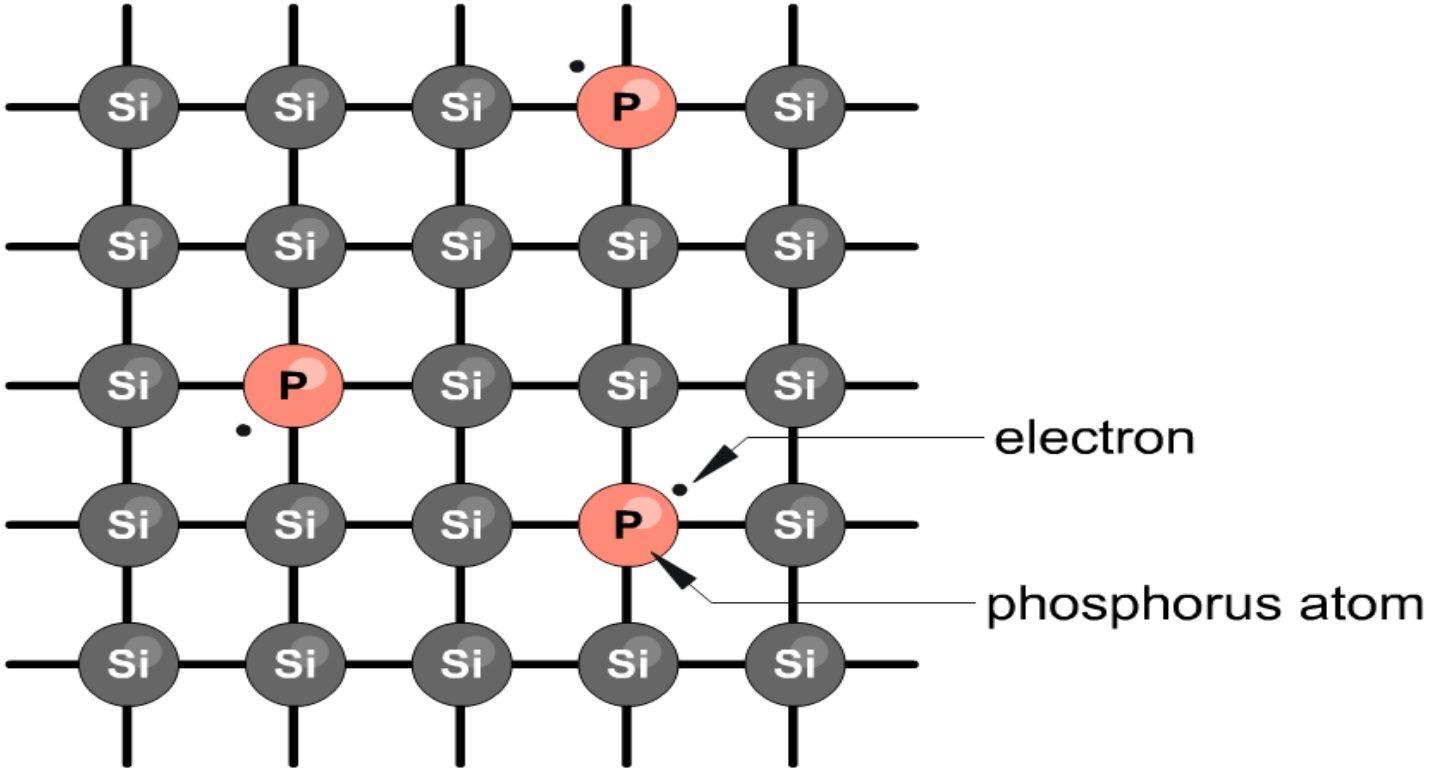
- N_c: Density of states in conduction band
- N_v: Density of states in valence band
- E_g: Band gap energy in eV
- k: Boltzmann constant 8.62×10^{-5} eV/K
- T: Temperature in K

For Silicon $E_g = 1.12$ eV $N_c = 3.2\times10^{19}$ cm^{-3} $N_v = 1.8\times10^{19}$ cm^{-3}

At 300 K $n_i = 1\times10^{10}$ cm^{-3}

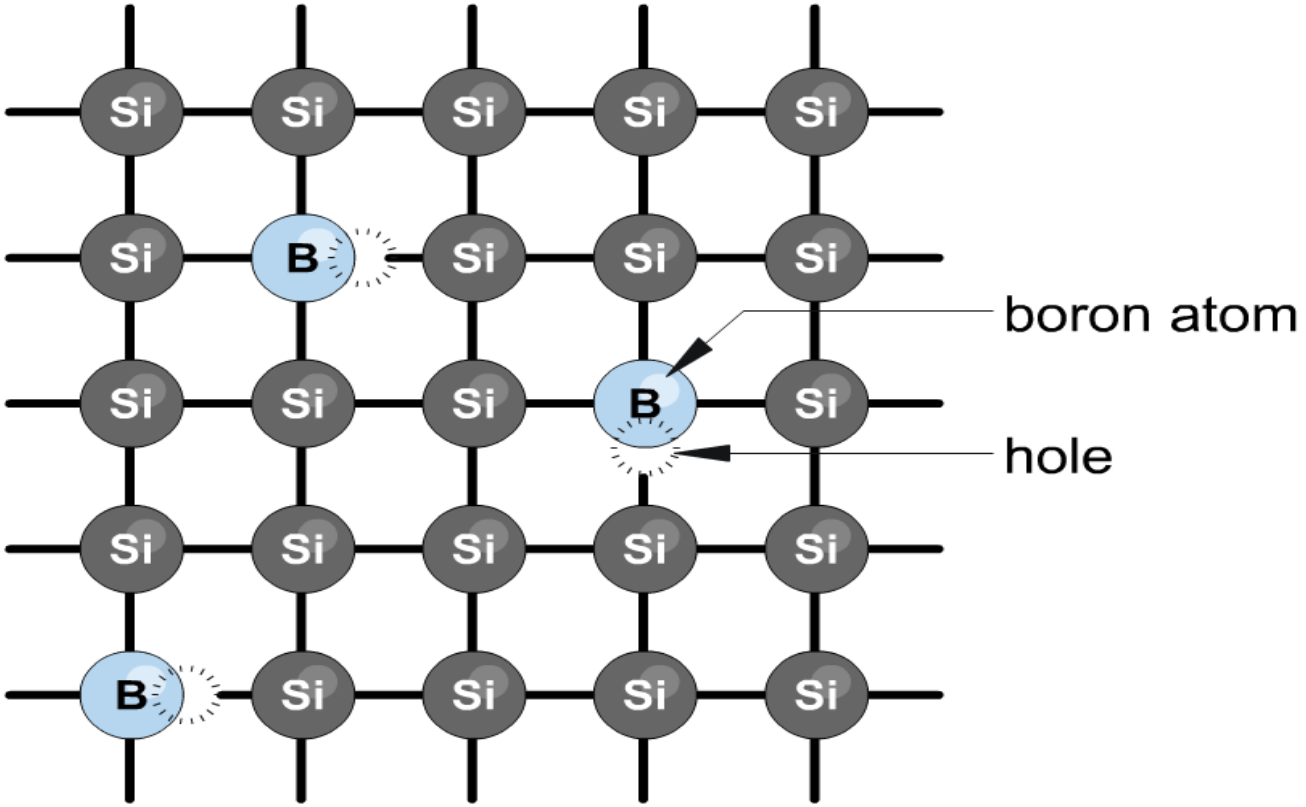
For a good conductor At 300 K $n_i = 1\times10^{23}$ cm^{-3}

Doping: N-type Si



Donor concentration (N_D) 10^{15}-10^{18} atom/cm^3

Doping: P-type Si



Acceptor concentration (N_A) 10^{13}-10^{16} atom/cm^3

Carrier Count: Doped Si

The density of free electrons is N-type silicon is given by

$$n \approx N_D$$

The density of holes electrons is P-type silicon is given by

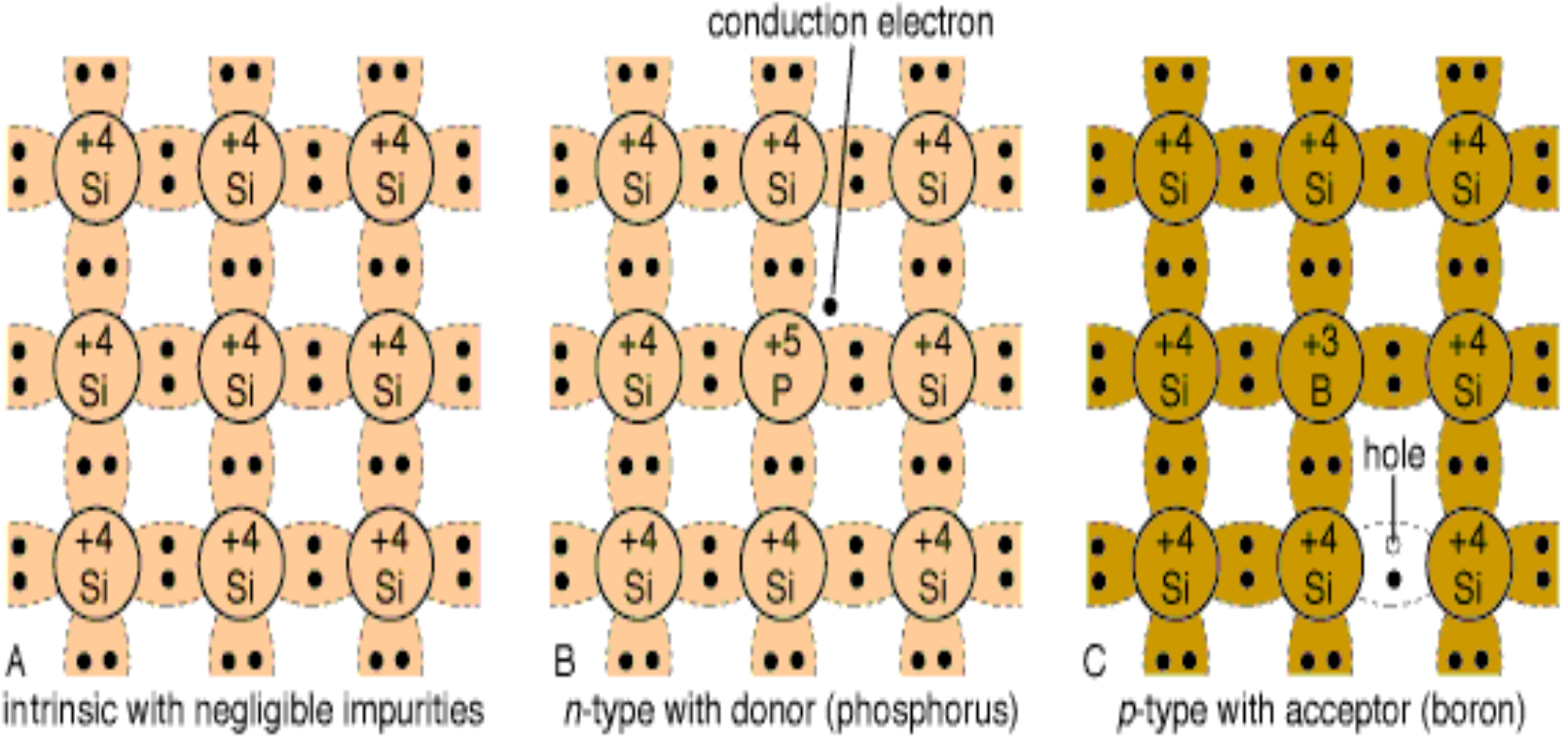
$$p \approx N_A$$

Does this doping change the original material?

Compare the number of original atoms to the number of acceptor/donor atoms.

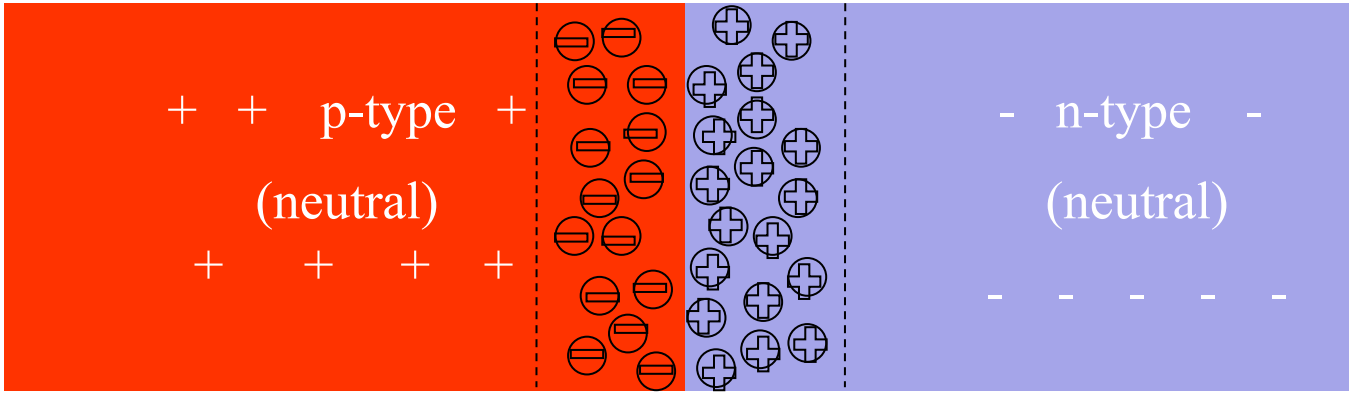
On average, the density of original atoms is about $10^{23} - 10^{24}$ atoms/cm^3.

Silicon Doping

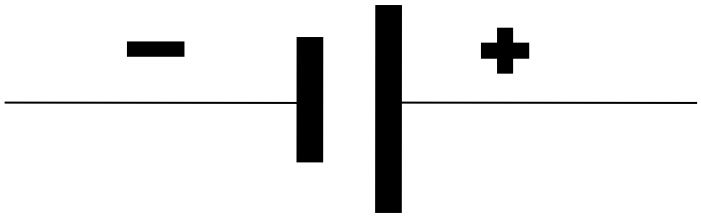


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The PN junction: The Diode

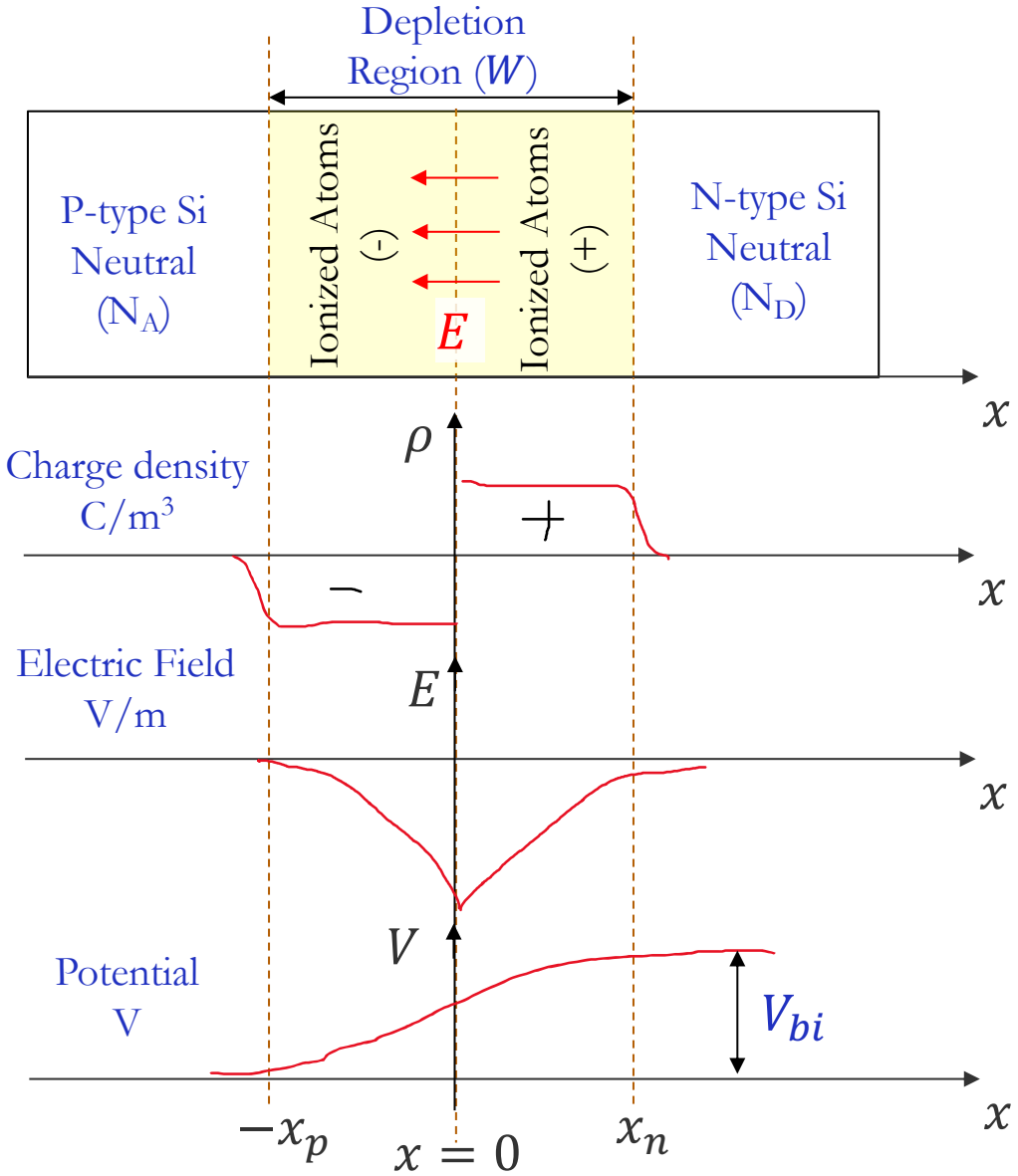


Built-in Voltage (V_{DO})





The PN Junction Parameters



$$V_{bi} = \frac{kT}{q}\ln\frac{N_D N_A}{n_i^2} \qquad W = x_n + x_p$$

$$x_n = \sqrt{\frac{2\varepsilon V_{bi}}{q}\frac{N_A}{N_D(N_A+N_D)}}$$

$$x_p = \sqrt{\frac{2\varepsilon V_{bi}}{q}\frac{N_D}{N_A(N_A+N_D)}}$$

$$E(x) = \frac{-qN_A}{\varepsilon}(x_p + x) \quad \text{p-side}$$

$$E(x) = \frac{-qN_D}{\varepsilon}(x_n - x) \quad \text{n-side}$$

$$V(x) = \frac{qN_A}{2\varepsilon}(x_p + x)^2 \quad \text{p-side}$$

$$V(x) = \frac{-qN_D}{2\varepsilon}(x_n - x)^2 + V_{bi} \quad \text{n-side}$$

$$C_j = \frac{\varepsilon A}{\sqrt{\frac{2\varepsilon(V_{bi}-V_A)}{q}\frac{N_A+N_D}{N_A N_D}}}$$

The PN Junction Parameters

Example (1): Consider a Si pn junction at room temperature with donor doping of $N_D = 10^{16}$ cm^{-3} and accepter doping of $N_A = 10^{15}$ cm^{-3}. Calculate junction width.

At 300 K $\quad n_i = 1\times10^{10}$ cm^{-3} $\quad V_{bi} = 0.633$ V

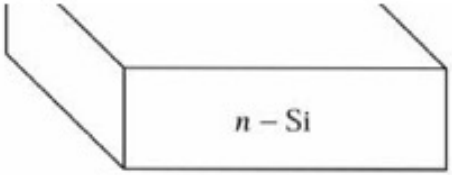
$x_p = 0.885\times10^{-6}$ m $\quad x_n = 0.0885\times10^{-6}$ m $\quad W = 0.973\ \mu$m

Example (2): For the junction in example 1, find the built-in voltage for $N_D = 10^{15}$ cm^{-3} and $N_D = 10^{17}$ cm^{-3}.

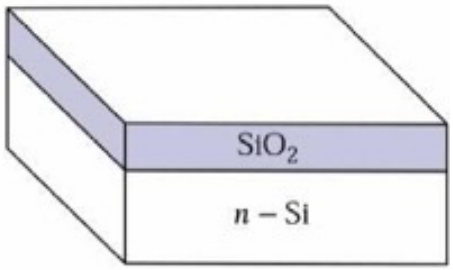
$$V_{bi} = 0.025 \ln \frac{N_D 10^{15}}{(10^{10})^2}$$

N_D (cm^{-3})	V_{bi} (V)
10^{15}	0.599
10^{16}	0.633
10^{17}	0.718

The PN junction: Fabrication

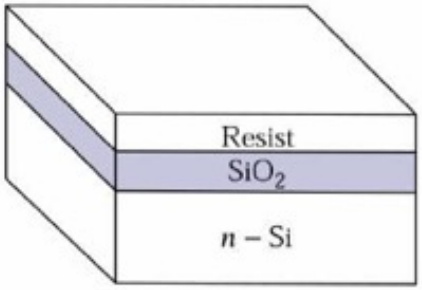


(a) Si wafer

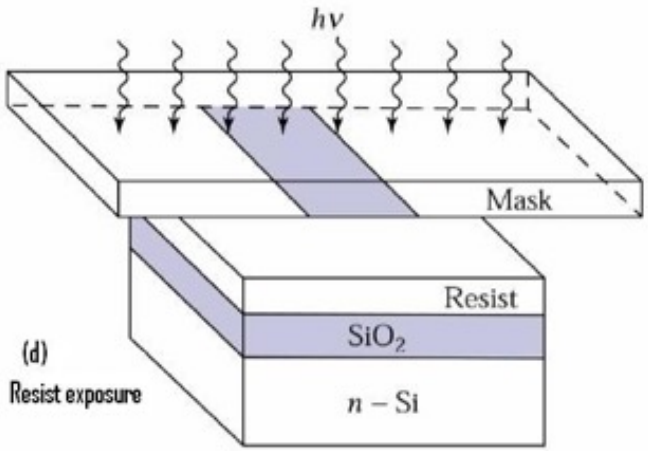


(b) Oxidation

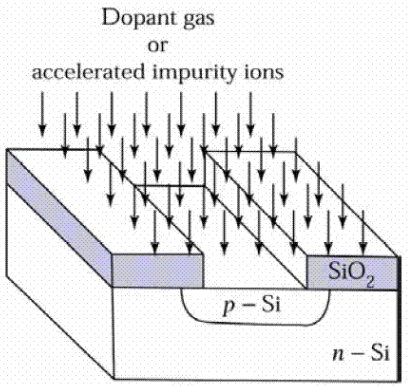




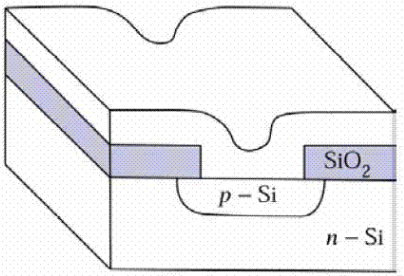
(c) Application of resist



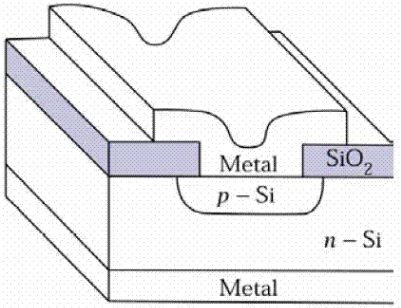
(d)
Resist exposure



(h) Diffusion or Implantation process



(i) Metallization



(j) Formed p-n junction

Q & A

Send me an email if
you have any
questions.